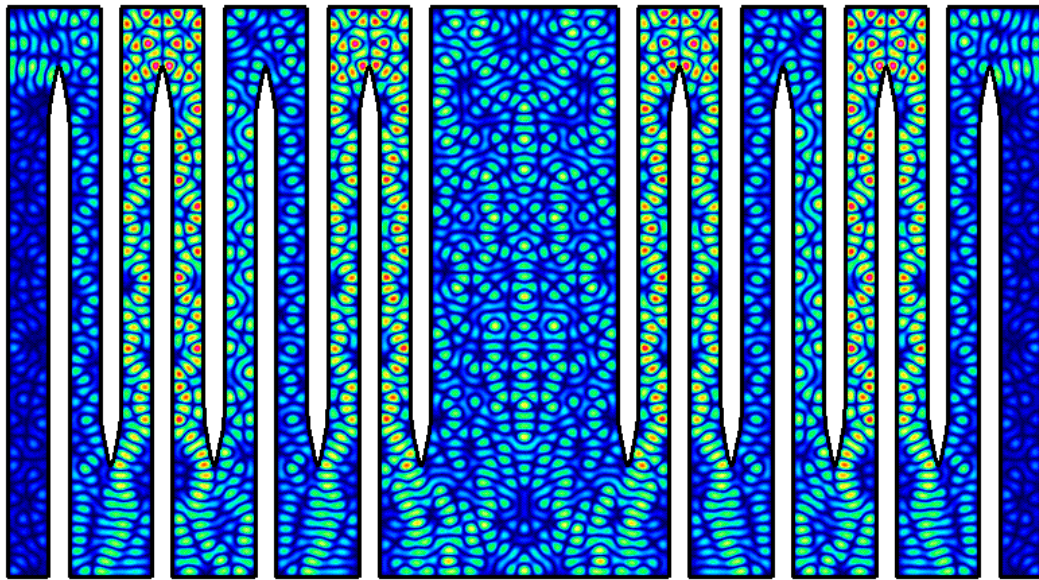


# Waves 2026 - Program

## The 17th International Conference on Mathematical and Numerical Aspects of Wave Propagation

John Molson school of business Concordia University in Montreal

Canada, June 22-26 2026



## **Monday - Tuesday**

**Registration open (08:00-)**

# Monday

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Welcome · 08:30-09:00

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## Plenary: Guillaume Bal

09:00-10:00 MB-1.210 · Session Chair: Chrysoula Tsogka

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Coffee 10:00-10:30

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## Block A · 10:30–12:30

MB-1.210 — Boundary Element Methods & Integral Equations #1 Session - Chair: Marc Bonnet

| Time  | Title                                                                                               | Authors                                                                                  |
|-------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 10:30 | Efficient evaluation of the two-layered acoustic Green's function                                   | <b>Stefan A. Sauter</b> , Laura Depedrini, Benedikt Grässle, J. Markus Melenk            |
| 11:00 | Fast high-order solvers for the Lippmann-Schwinger equation in piecewise-smooth heterogeneous media | <b>Thomas G. Anderson</b> , Juan Burbano-Gallegos, Luiz M. Faria, Carlos Pérez-Arancibia |
| 11:30 | A fast direct solver based on FEM-BEM coupling for time-harmonic electromagnetic scattering         | <b>Johann Bourhis</b> , Emanuele Arcese, Francis Collino                                 |
| 12:00 | Truncated kernel windowed Fourier projection: a fast algorithm for the 3D free-space wave equation  | <b>Nour G. Al Hassanieh</b> , Alex H. Barnett, Leslie Greengard                          |

MB-9.A — Helmholtz #1 Session - Chair: Joseph Coyle

| Time  | Title                                                                              | Authors                                          |
|-------|------------------------------------------------------------------------------------|--------------------------------------------------|
| 10:30 | Convergence of the WaveHoltz iteration on $\mathbb{R}^d$                           | <b>Elliot Backman</b> , Olof Runborg             |
| 11:00 | Effective models for Helmholtz's equation in a corner with a thin layer            | <b>Cédric Baudet</b> , Sonia Fliss, Patrick Joly |
| 11:30 | The largest ever Helmholtz simulations in inhomogeneous media                      | <b>Max Cubillos</b> , Edwin Jimenez              |
| 12:00 | Nonlinear acoustic inverse problem with cross-correlation data for passive imaging | <b>Jean Dutheil</b> , Florian Faucher            |

**MB-9.B — Acoustic Wave Propagation & Scattering #1 Session - Chair: Christophe Hazard**

| Time  | Title                                                                             | Authors                                                            |
|-------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 10:30 | Time dependent acoustic scattering from penetrable objects with nonlinearities    | <b>Paschalis Angelidis</b> , Alexander Rieder                      |
| 11:00 | Spectral analysis of a closed perturbed waveguide filled with a hyperbolic medium | Maryna Kachanovska, <b>Dylan Machado</b>                           |
| 11:30 | Rendering disordered waveguides transparent to waves                              | <b>Mathieu Davy</b> , Isam Ben Soltane, Daniel Lenz, Stefan Rotter |
| 12:00 | Spectral properties of a kinetic equation coupled with the linear wave system     | <b>Bruno Després</b>                                               |

**MB-9.C — Inverse Problems & Imaging #1 - Session Chair: Daniel Appelo**

| Time  | Title                                                                                                 | Authors                                                                            |
|-------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 10:30 | Recovery of acoustic properties in the Westervelt equation using high-frequency large-amplitude waves | <b>Sebastian Acosta</b> , Benjamin Palacios                                        |
| 11:00 | A neural operator framework for solving inverse scattering problems                                   | <b>Victor Chenu</b> , Housseem Haddar, Hadrien Montanelli                          |
| 11:30 | A deep learning approach for optical cloaking device design                                           | <b>Elsie Cortes</b> , Camille Carvalho, Symeon Papadimitropoulos, Chrysoula Tsogka |
| 12:00 | A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media             | <b>General Ozochiawaeze</b> , Rafael Ceja Ayala, Isaac Harris                      |

**MB-9.D — Absorbing Boundary Conditions #1 - Session Chair: Maryna Kachanovska**

| Time  | Title                                                                                                           | Authors                                                                                            |
|-------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 10:30 | Revisiting Artificial Boundary Conditions Through Modal Expansion                                               | <b>Hélène Barucq</b> , Florian Faucher, Olivier Lafitte, Ha Pham                                   |
| 11:00 | Transparent boundary conditions in heterogeneous electromagnetic waveguides                                     | <b>Anne-Sophie Bonnet-Ben Dhia</b> , Lucas Chesnel, Sonia Fliss, Eric Lunéville, Aurélien Parigaux |
| 11:30 | Domain Truncation for Quasilinear Wave Equations                                                                | <b>Thomas Hagstrom</b>                                                                             |
| 12:00 | Finite element discretizations of hyperboloidal compactification layers for unbounded time-domain wave problems | <b>Markus Wess</b> , Anil Zenginoğlu                                                               |

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**Lunch 12:30-14:00**

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## Plenary: Christophe Hazard

14:00-15:00 MB-1.210 · Session Chair: Anne-Sophie Bonnet-Ben Dhia

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## Coffee 15:00-15:30

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## Block B · 15:30–17:30

**MB-1.210 — Elastic Waves & Elastodynamics #1 - Session Chair: Stephanie Chaillat**

| Time  | Title                                                                                               | Authors                                                    |
|-------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 15:30 | A Robust Model Order Reduction Method to Accelerate Full Waveform Inversion via Fréchet Derivatives | <b>Julien Besset</b> , Hélène Barucq, Victor Martins Gomes |
| 16:00 | Green tensor for strain gradient elastodynamics                                                     | <b>Marc Bonnet</b>                                         |
| 16:30 | Reconstruction of the Shear Modulus in Nearly Incompressible Media via Adaptive Spectral Inversion  | <b>Nagham Chibli</b> , Martin Genet, Sébastien Imperiale   |
| 17:00 | Spectral properties of waves in high-contrast media                                                 | Yuri A. Godin, Leonid Koralov, <b>Boris Vainberg</b>       |

**MB-9.A — Electromagnetic Waves & Maxwell Equations #1 - Session Chair: Carlos Perez-Arancibia**

| Time  | Title                                                                                                       | Authors                                                         |
|-------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 15:30 | Optimization of EM-Chiral Scatterers                                                                        | <b>Tilo Arens</b> , Marvin Knöller, Raphael Schurr              |
| 16:00 | Domain derivative based shape reconstruction for time-dependent Maxwell's equations                         | <b>Marvin Knöller</b>                                           |
| 16:30 | Towards fast conservative methods for the numerical resolution of Maxwell's equations in multipatch domains | <b>Ángel Pita-da-Veiga</b> , Jerónimo Rodríguez, Rafael Vázquez |
| 17:00 | The Hermite-Taylor Discrete Correction Function Method for Surface-Driven Electromagnetic Problems          | <b>Yann-Meing Law</b>                                           |

**MB-9.B — Helioseismology & Astrophysics Session Chair: Philippe Moireau**

| Time  | Title                                                                                    | Authors                                                                                         |
|-------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 15:30 | Reconstructing inertial-mode flows from the correlation function of acoustic modes       | Hélène Barucq, Florian Faucher, <b>Damien Fournier</b> , Laurent Gizon, Zhi-Chao Liang, Ha Pham |
| 16:00 | Uniqueness results in time-harmonic passive imaging with applications to helioseismology | <b>Thorsten Hohage</b> , Björn Müller                                                           |
| 16:30 | Asymptotic Solutions to the Radial Wave Equation in Kerr Spacetime                       | <b>John Levis</b> , Thomas Hagstrom                                                             |
| 17:00 | Stability of a passive inverse parameter problem with applications to helioseismology    | <b>Björn Müller</b> , Thorsten Hohage                                                           |

**MB-9.C — Nonlinear Waves, Solitons & Dynamical Systems - Session Chair: Behrooz Yousefzadeh**

| Time  | Title                                                                                         | Authors                                                                     |
|-------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 15:30 | Localized Time-Periodic Patterns in a 1D Lattice                                              | <b>Jason J. Bramburger</b> , Eric Berglund, Björn Sandstede                 |
| 16:00 | Bistability of travelling waves and mesa states in a mass-conserved reaction-diffusion system | <b>Jack M. Hughes</b> , Saar O. Modai, Leah Edelstein-Keshet, Arik Yochelis |
| 16:30 | Leafnet algorithm for N-dimensional nonlinear hyperbolic systems                              | <b>Emmanuel Lorin</b> , Arian Novruzi                                       |
| 17:00 | Exploring Explicit Formulas and Multi-solitons for Integrable Systems                         | <b>Ruoci Sun</b>                                                            |

**MB-9.D — Water Waves & Wave-Structure Interaction #1 - Session Chair: Olivier Lafitte**

| Time  | Title                                                                                                      | Authors                                                       |
|-------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 15:30 | Continuous and discrete stability of dispersive perturbation of hyperbolic initial boundary value problems | <b>Geoffrey Beck</b> , Nicolas Crouseilles, Ludovic Martaud   |
| 16:00 | The return to the equilibrium of a floating cylinder in the Boussinesq regime                              | Geoffrey Beck, <b>Ewan Contentin</b> , Ludovic Martaud        |
| 16:30 | A Theory for Three-Dimensional Wave Breaking Based on Crest Instability of Short-Crested Stokes Waves      | <b>Frédéric Dias</b> , Ayoub Mansar, Matt Turner, Tom Bridges |
| 17:00 | Wave-Based Stability Analysis of Couette and Gravity-Driven Bilayer Film Flows                             | <b>M. Al Ahmadi Hammou</b> , M. Scholle, P. H. Gaskell        |

## Tuesday

### Plenary: Barbara Romanowicz

09:00-10:00 MB-1.210 · Session Chair: Serge Prudhomme

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### Coffee 10:00-10:30

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### Block A · 10:30–12:30

**MB-1.210 — Helmholtz #2 - Session Chair: Thomas Anderson**

| Time  | Title                                                                                          | Authors                                                             |
|-------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 10:30 | A Helmholtz Solver for Open Domains with Complex Geometry using WaveHoltz and Overset Grids    | Jeffrey Banks, William Henshaw, <b>Ngan Le</b> , Donald Schwendeman |
| 11:00 | A Trefftz Continuous Galerkin method for Helmholtz problems                                    | <b>Nicola Galante</b> , Bruno Després, Emile Parolin                |
| 11:30 | High-order frequency derivatives of Helmholtz BIE solutions: fast computation and applications | <b>Hiroshi Isakari</b>                                              |
| 12:00 | Accelerating WaveHoltz with Deflation                                                          | <b>Daniel Appelo</b> , William Henshaw                              |

**MB-9.A — Boundary Element Methods & Integral Equations #2 - Session Chair: Stefan Sauter**

| Time  | Title                                                                                                   | Authors                                                                                                              |
|-------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 10:30 | Efficient 3D elastodynamic multiple scattering using the Method of Reflections                          | <b>Stéphanie Chaillat</b> , Marion Darbas                                                                            |
| 11:00 | Fast Singular-Kernel Convolution on General Non-Smooth Domains via Truncated Fourier Filtering          | <b>Jinghao Cao</b> , Oscar Bruno                                                                                     |
| 11:30 | Convergence analysis for Numerical Steepest Descent contour deformation methods                         | <b>Thomas Caussade</b> , David P. Hewett                                                                             |
| 12:00 | Space-time BEM based coupling strategies for the analysis of infrastructures solicited by seismic waves | <b>Giulia Di Credico</b> , Alessandra Aimi, Heiko Gimperlein, Chiara Guardasoni, Francesco Freddi, Roberto Valentino |

**MB-9.B — Acoustic Wave Propagation & Scattering #2 - Session Chair: Frederic Dias**

| Time  | Title                                                                          | Authors                                                                             |
|-------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 10:30 | Scattering problems in junctions of stratified media                           | Sarah Al Humaikani, Anne-Sophie Bonnet-Ben Dhia, <b>Sonia Fliss</b>                 |
| 11:00 | Analysis of the geometric spectral properties of electromagnetic waveguides    | Philippe Briet, <b>Maxence Cassier</b> , Thomas Ourmières-Bonafos, Michele Zaccaron |
| 11:30 | Exceptional Points and Defective Resonances in a Sound-Soft Scattering Problem | <b>Kei Matsushima</b>                                                               |
| 12:00 | Flow Rate Measurement in a Heterogeneous Fluid with Acoustic Waves             | <b>Alexander McSweeney-Davis</b> , Lorenzo Audibert, Housseem Haddar                |

**MB-9.C — Inverse Problems & Imaging #2 - Session Chair: Max Cubillos**

| Time  | Title                                                                         | Authors                                                                                                |
|-------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 10:30 | Explicit inverse scattering for the one-dimensional Schrödinger equation      | <b>Peter Gibson</b>                                                                                    |
| 11:00 | On the role of regularisation parameters in Adaptive Eigenspace Inversion     | <b>Marie Graff</b> , Nasrin Nikbakht, Melissa Tacy                                                     |
| 11:30 | The Inverse Elasto-Acoustic Problem                                           | <b>Patrick Grice</b> , Yassine Boubendir, Zoi-Heleni Michalopoulou                                     |
| 12:00 | Adapting Full-Waveform Inversion for Subsurface CO <sub>2</sub> Sequestration | <b>Alison Malcolm</b> , Vahid Entezar-Saadat, Xingda Jiang, Abolfazl Khan Mohammadi, Colin Farquharson |

**MB-9.D — Absorbing Boundary Conditions #2 - Session chair: Thorsten Hohage**

| Time  | Title                                                                                                           | Authors                                                                            |
|-------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 10:30 | Transparent boundary conditions for the stellar oscillation equations without Cowling approximation             | <b>Janosch Preuss</b> , Hélène Barucq, Florian Faucher, Ha Pham, Sébastien Tordeux |
| 11:00 | Interfacial Radiation Boundary Conditions for Topologically Protected Edge Modes                                | <b>Mason McCallum</b> , Thomas Hagstrom                                            |
| 11:30 | Complex scaling for the homogeneous wave equation with scaling direction non-normal to the transparent boundary | <b>Lothar Nannen</b> , Constantin Pierer, Markus Wess                              |
| 12:00 | A potential theory on weighted graphs                                                                           | <b>Fernando Guevara Vasquez</b> , Trent DeGiovanni                                 |



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## Lunch 12:30-14:00

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## Plenary: Isaac Harris

14:00-15:00 MB-1.210 · Session Chair: Behrooz Yousefzadeh

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## Coffee 15:00-15:30

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## Tue · Block B · 15:30–17:30

**MB-1.210 — Elastic Waves & Elastodynamics #2 - Session Chair: Barbara Romanowicz**

| Time  | Title                                                                                              | Authors                                                                                         |
|-------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 15:30 | Comparison of numerical coupling strategies in time-harmonic elasto-acoustic simulations           | <b>Pierre Dubois</b> , H       Barucq, Florian Faucher, Isabelle Rami      , Rapha       Prat   |
| 16:00 | A well-posed effective strain-gradient model for elastic waves in arbitrary periodic media         | <b>Nicolas Auffray</b> , Marc Bonnet, R       Cornaggia, Giuseppe Rosi                          |
| 16:30 | Mathematical models for wave propagation phenomena generated by landslides                         | Juliette Dubois, Natacha Guegan-Fau, <b>S            </b> , Anne Mangeney, Jacques Sainte-Marie |
| 17:00 | High-order Simulation of Elasto-Acoustic Wave Propagation Using Hybrid Nonconforming Methods (HHO) | <b>Romain Mottier</b> , Alexandre Ern, Laurent Guillot                                          |

**MB-9.A — Electromagnetic Waves & Maxwell Equations #2 - Session Chair: Tilo Arens**

| Time  | Title                                                                                                   | Authors                                                        |
|-------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 15:30 | A Stochastic Boundary Element Method for Electromagnetic Scattering in Random Media                     | <b>Titouan Marquaille</b> , Thomas Bonnafont, Arnaud Coatanhay |
| 16:00 | A Discontinuous Petrov–Galerkin Framework for Time-Harmonic Maxwell Equations in Microwave-Heated Flows | <b>Oreste Marquis</b> , Bruno Blais, Matthias Maier            |
| 16:30 | Computational spectral geometry for the Maxwell cavity problem                                          | <b>Nilima Nigam</b> , Kewen Bu, Jayda Grant, Rodrigo Stehling  |
| 17:00 | A Hermite-Taylor Compact Discrete Correction Function Method for Maxwell's Equations                    | <b>Seth Malonson</b> , Yann-Meing Law, Jean-Christophe Nave    |

**MB-9.C — Stochastic Propagation - Session Chair: Emmanuel Lorin**

| Time  | Title                                                                                       | Authors                                                           |
|-------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 15:30 | Effective attenuation and Gaussian statistics of the speckle for the paraxial wave equation | <b>Christophe Gomez</b> , Olivier Pinaud                          |
| 16:00 | Multiscale analysis of wave propagation in the branched flow regime                         | Josselin Garnier, <b>Natacha Guegan-Fau</b> , Sébastien Imperiale |
| 16:30 | Optical wave reconstruction in highly turbulent random media                                | <b>Kevin Luna</b> , Max Cubillos                                  |
| 17:00 | Destabilization of traveling waves in a nonlinear lattice with nonreciprocal coupling       | Andrus Giraldo, Stefan Ruschel, <b>Behrooz Yousefzadeh</b>        |

**MB-9.B — Periodic Media, Metamaterials & Homogenization #1 - Session Chair: Olof Runborg**

| Time  | Title                                                                                          | Authors                                                                |
|-------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 15:30 | Diffraction by a rough thin layer on an arbitrary shaped object: the periodic and random cases | <b>Pierre Boulogne</b> , Sonia Fliss, Laure Giovangigli, Justine Labat |
| 16:00 | High-order stochastic homogenization of the Helmholtz transmission problem                     | <b>Laure Giovangigli</b>                                               |
| 16:30 | Wave propagation and gaps in three-dimensional periodic media                                  | <b>Yuri Godin</b> , Boris Vainberg                                     |
| 17:00 | On the surface Bloch waves in truncated periodic media                                         | <b>Bojan B Guzina</b> , Shixu Meng, Prasanna Salasiya, Long Nguyen     |

**MB-9.D — Time-Domain Methods & A Posteriori Estimation #1 - Session Chair: Marcus Grote**

| Time  | Title                                                                                | Authors                                                          |
|-------|--------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 15:30 | A note on the discrete Unmapped Tent Pitching for the heterogeneous wave equation    | <b>Marcella Bonazzoli</b> , Gabriele Ciaramella, Ilario Mazzieri |
| 16:00 | Decoupling Wave-Heat-Type Problems                                                   | <b>Julian Dörner</b> , Benjamin Dörich                           |
| 16:30 | Mixed Precision Implicit Runge–Kutta and Explicit Two-Derivative Runge–Kutta Methods | <b>Sigal Gottlieb</b> , Zachary J. Grant, César Herrera          |
| 17:00 | Recovery-Based Error Indicator for Finite Difference Methods                         | <b>Ferhat Sindy</b> , Annalisa Buffa, Marco Picasso              |

**Reception - 17:30-20:30**

# Wednesday

Social day

## Thursday

### Plenary: Andrea Moiola

09:00-10:00 MB-1.210 · Session Chair Lise-Marie Imbert-Gerard

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### Coffee 10:00-10:30

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### Thu · Block A · 10:30–12:30

MB-1.210 — Helmholtz #3 - Session Chair: Sonia Fliss

| Time  | Title                                                                                                                          | Authors                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 10:30 | The hp-FEM Does Not Suffer from the Pollution Effect When Applied to the Helmholtz Equation with Piecewise Gevrey Coefficients | Jeffrey Galkowski, <b>Mostafa Meliani</b> , Euan Spence                        |
| 11:00 | High-order Wavelet Galerkin Methods for Scattering Problems                                                                    | <b>Michelle Michelle</b> , Bin Han                                             |
| 11:30 | A HDG method with transmission variables for time-harmonic wave propagation problems                                           | <b>Simone Pescuma</b> , Gwénaél Gabard, Théophile Chaumont-Frelet, Axel Modave |
| 12:00 | Scalable WaveHoltz in MFEM                                                                                                     | <b>Amit Rotem</b> , Daniel Appelö, Will Pazner                                 |

MB-9.A — Boundary Element Methods & Integral Equations #3 - Session Chair: Johannes Tausch

| Time  | Title                                                                                                                            | Authors                                                 |
|-------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 10:30 | A posteriori estimates and adaptive boundary elements for the wave equation                                                      | <b>Heiko Gimperlein</b>                                 |
| 11:00 | Towards kernel-free boundary integral methods for variable coefficients                                                          | <b>Benedikt Gräble</b> , Stefan A. Sauter               |
| 11:30 | Geometry-conforming Galerkin methods for the Lippmann-Schwinger equation on fractal domains                                      | <b>David P. Hewett</b> , Joshua Bannister, Andrew Gibbs |
| 12:00 | An Unconditionally Stable Energetic Galerkin Scheme for Acoustic Penetrable Objects Using Time-Domain Surface Integral Equations | <b>Junze Shao</b> , Rui Chen, Hakan Bagci               |

**MB-9.B — Acoustic Wave Propagation & Scattering #3 - Session Chair: Lothar Nannen**

| Time  | Title                                                                                                                      | Authors                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 10:30 | Imaging in two-dimensional waveguides with complex geometry                                                                | <b>Gregory Mwamba</b> , Chrysoula Tsogka, Symeon Papadimitropoulos                                           |
| 11:00 | The cut-off resolvent can grow arbitrarily fast in obstacle scattering                                                     | <b>Siavash Sadeghi</b> , Simon N. Chandler-Wilde                                                             |
| 11:30 | Application of the multivariate Loewner Framework to the modelling of acoustic propagation in rigid-frame porous materials | <b>Lucie Schwoebel</b> , Charles Poussot-Vassal, Denis Matignon, Rémi Roncen, Estelle Piot, Pierre Vuillemin |
| 12:00 | Optimization-based transionospheric SAR autofocus with screen projection precursor                                         | Mikhail Gilman, <b>Semyon Tsynkov</b>                                                                        |

**MB-9.C — Inverse Problems & Imaging #3 - Session Chair: Isaac Harris**

| Time  | Title                                                                                                                            | Authors                                                                           |
|-------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 10:30 | Transmission eigenvalues for biharmonic plate scattering: existence, discreteness, far-field recovery, and numerical computation | <b>Andreas Kleefeld</b> , Isaac Harris, Heejin Lee                                |
| 11:00 | Characterizing the Imaging Indicator of the Linear Sampling Method for the Inverse Medium Scattering Problem                     | Lorenzo Audibert, <b>Shixu Meng</b>                                               |
| 11:30 | Spectrally characterizing subsurface targets with GPR-SAR                                                                        | Arnold D. Kim, Symeon Papadimitropoulos, <b>Joseph Simpson</b> , Chrysoula Tsogka |
| 12:00 | Identifying defective units in infinite periodic arrays of point sources                                                         | <b>Dinh-Liem Nguyen</b> , Nhung H. Nguyen, Thi-Phong Nguyen                       |

**MB-9.D — Trefftz & Plane Wave Methods - Session Chair: Bruno Despres**

| Time  | Title                                                                  | Authors                                          |
|-------|------------------------------------------------------------------------|--------------------------------------------------|
| 10:30 | A Model Reduction Approach for Wave Phenomena using dG-Trefftz Methods | <b>Tobias Born</b> , Karsten Urban               |
| 11:00 | Preconditioning for the DG plane wave method                           | <b>Joseph Coyle</b> , Nilima Nigam               |
| 11:30 | Polynomial quasi-Trefftz spaces for scalar linear equations            | <b>Lise-Marie Imbert-Gérard</b>                  |
| 12:00 | Quasi-Trefftz spaces for time-harmonic Maxwell's equations             | <b>Ilaria Fontana</b> , Lise-Marie Imbert-Gérard |

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**Lunch 12:30-14:00**

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## Plenary: Lilia Krivodonova

14:00 – 15:00 MB-1.210 · Session Chair: Tom Hagstrom

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## Coffee 15:00-15:30

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### Thu · Block B · 15:30 – 17:30

MB-1.210 — Periodic Media, Metamaterials & Homogenization #2 - Session Chair: Sebastien

Imperiale

| Time  | Title                                                                           | Authors                                               |
|-------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| 15:30 | Analysis of Wave Propagation in Time-Dependent Media                            | <b>Patrick Joly</b> , Marie Touboul                   |
| 16:00 | Homogenization of wave equations and dispersive profiles for ring solutions     | <b>Ben Schweizer</b> , Grégoire Allaire, Agnes Lamacz |
| 16:30 | Radiation Boundary Conditions with Rational Function Approximations             | <b>Steve Li</b> , Thomas Hagstrom                     |
| 17:00 | Error Estimation for a Homogenized Model of a Space-Time Modulated Metamaterial | <b>Jean-Francois Mercier</b> , Marie Touboul          |

MB-9.A — Time-Domain Methods & A Posteriori Estimation #2 - Session Chair: Shixu Meng

| Time  | Title                                                                                            | Authors                                                     |
|-------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 15:30 | Adaptive FEM with explicit time integration for the wave equation                                | <b>Marcus J. Grote</b> , Omar Lakkis, Carina S. Santos      |
| 16:00 | High-order contour-integral methods for strongly continuous semigroups                           | <b>Andrew Horning</b> , Adam R. Gerlach                     |
| 16:30 | Reliable and locally space-time efficient a posteriori estimates for the transient wave equation | <b>Nicolas Hugot</b> , Alexandre Imperiale, Martin Vohralík |
| 17:00 | A parallel space-time p-adaptive discontinuous Galerkin method for nonlinear acoustics           | <b>Pascal Lehner</b> , Christian Wieners, Danielle Corallo  |

**MB-9.B — Water Waves & Wave-Structure Interaction #2 - Session Chair: Helene Barucq**

| Time  | Title                                                                               | Authors                                                                                                        |
|-------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 15:30 | Waves and transport in a circular water wave resonator: experiments and simulations | <b>Uwe Harlander</b> , Franz-Theo Schön, Ion D. Borcia, Rodica Borcia, Seymen Ilke Kaykanat, Michael Bestehorn |
| 16:00 | Propagation of the heave in a harbor: geometrical optics approach                   | <b>Olivier Lafitte</b> , Emmanuel Audusse, Catherine Sulem                                                     |
| 16:30 | Interactions between water waves and floating structures, a numerical method        | Geoffrey Beck, <b>Alan Riquier</b>                                                                             |
| 17:00 | Viscoelastic gravity wave force on a circular cylinder in open channel              | <b>Boyuan Yu</b> , Vincent H. Chu                                                                              |

**MB-9.C — Helmholtz / Trefftz / BEM Solver Extensions - Session Chair: Andrea Moiola**

| Time  | Title                                                                                                               | Authors                                                                                  |
|-------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 15:30 | Infinity as a finite boundary for Helmholtz equations                                                               | <b>Anil Zenginoğlu</b> , Markus Wess                                                     |
| 16:00 | Quasi-Trefftz Discontinuous Galerkin Method for the Convected Helmholtz Equation with Smooth Coefficients           | <b>Andréa Lagardère</b> , Lise-Marie Imbert-Gerard, Guillaume Sylvand, Sébastien Tordeux |
| 16:30 | Variational Quasi-Trefftz Method for Time-Harmonic First-Order Wave Propagation Systems                             | Matthias Rivet, Sébastien Pernet, <b>Sébastien Tordeux</b>                               |
| 17:00 | Fast 3D Time Domain Boundary Element Approach for Elastodynamics based on Multivariate Adaptive Cross Approximation | Vibudha Lakshmi Keshava, <b>Martin Schanz</b>                                            |

**MB-9.D — Electromagnetic Waves & Maxwell Equations #3 - Session Chair: Daniel Appelo**

| Time  | Title                                                                                                    | Authors                                                                  |
|-------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 15:30 | Broken-FEEC on multipatch domains with local refinements                                                 | <b>Frederik Schnack</b> , Martin Campos Pinto                            |
| 16:00 | Topological solutions and accelerated waves in electromagnetic theory                                    | Radmila Sazdanovic, Paul Spears, <b>Semyon Tsynkov</b>                   |
| 16:30 | Energy-preserving simulation of electromagnetic waves at oblique incidence using Split-Field FETD method | <b>Valentin Wasquel</b> , Paula Aguilera, Michel Fournié, Denis Matignon |
| 17:00 | Derivation of polarization dependent interface conditions for Maxwell's equations                        | <b>David Wiedemann</b> , Ben Schweizer, Bérangère Delourme               |

**Conference Banquet 18:00 -**

## Friday

### Plenary: Leonardo Zepeda-Núñez

09:00-10:00 MB-1.210 · Session Chair: Daniel Appelo

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### Coffee 10:00-10:30

### Fri · Block A · 10:30 – 12:30

**MB-1.210 — Integral / DD / ABC - Session Chair: Nilima Nigam**

| Time  | Title                                                                               | Authors                                                                                                                   |
|-------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 10:30 | Quadrature for Space-Time Galerkin BEM of the Wave Equation                         | <b>Johannes Tausch</b>                                                                                                    |
| 11:00 | Transcranial Ultrasound Simulations using Volume-Surface Integral Equations         | <b>Elwin van 't Wout</b> , Alberto Almuna-Morales, Danilo Aballay, Ignacio Labarca-Figueroa, Pierre Gélât, Reza Haqshenas |
| 11:30 | Optimized Schwarz methods in time for discrete transport control                    | Duc Quang Bui, Bérangère Delourme, Laurence Halpern, <b>Felix Kwok</b>                                                    |
| 12:00 | A Multi-domain/Multi-method Preconditioner for the Electric Field Integral Equation | <b>Timothée Galtier</b> , David Levadoux                                                                                  |

**MB-9.A — Boundary Element Methods & Integral Equations #4 - Session Chair: David Hewitt**

| Time  | Title                                                                                                          | Authors                                                                                                                 |
|-------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 10:30 | Maxwell à la Helmholtz: Frequency-Robust Direct Boundary Integral Equations for 3D PEC Scattering              | <b>Carlos Pérez-Arancibia</b> , Catalin Turc                                                                            |
| 11:00 | Fast assembly of H-matrix and FMM-based boundary operators for time-harmonic Maxwell equations                 | <b>Clément Aumonier</b> , David Levadoux                                                                                |
| 11:30 | High-order Nyström method for scattering by penetrable inhomogeneous media with discontinuous refractive index | <b>Ambuj Pandey</b> , Krishna Yamanappa Poojara                                                                         |
| 12:00 | Model reduction for a wave propagation problem in a slotted structure                                          | <b>Augustin Leclerc</b> , Pierre Benjamin, Luc Giraud, Sofiane Haddad, Paul Mycek, Guillaume Sylvand, Isabelle Terrasse |



**MB-9.B — Acoustic Wave Propagation & Scattering #4 - Session Chair: Lilia Krivodonova**

| Time  | Title                                                                                                                         | Authors                                                                                        |
|-------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 10:30 | Passivity enforcing multipole based method for time domain modelling of acoustic wave propagation in rigid frame porous media | <b>Lucie Schwoebel</b> , Denis Matignon, Remi Roncen, Estelle Piot                             |
| 11:00 | Quantitative ultrasound imaging in layered media                                                                              | Josselin Garnier, Laure Giovangigli, Pierre Millien, <b>Sofia Suarez Casabiell</b>             |
| 11:30 | Addressing Seismic Acquisition Repeatability in CO <sub>2</sub> Storage Monitoring Using Machine Learning                     | <b>Manon Sarrouilhe</b> , Mauricio Araya-Polo, Hélène Barucq, Henri Calandra, Stefano Frambati |
| 12:00 | Towards a Scale-Bridging Time Discretization Scheme for the Acoustic Wave Equation                                            | <b>Dustin Mühlhäuser</b> , Roland Maier                                                        |

**MB-9.C — Inverse Problems & Imaging #4 - Session Chair: Andreas Kleefeld**

| Time  | Title                                                                                           | Authors                                                                     |
|-------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 10:30 | Some approaches for the field extrapolation in the context of inverse magnetisation problem     | <b>Dmitry Ponomarev</b>                                                     |
| 11:00 | A tsunami inverse problem in the time domain: numerical analysis of an optimal control approach | Laurent Bourgeois, <b>Philippe Moireau</b> , Raphaël Terrine                |
| 11:30 | Inversion-free binary segmentation for computed tomography and inverse scattering               | <b>Jacob D. Rezac</b> , Zachary J. Grey, Newell H. Moser, Jason P. Killgore |
| 12:00 | Direct reconstruction for acoustic inverse Born scattering                                      | <b>Lisa Schätzle</b> , Nuutti Hyvönen                                       |

**MB-9.D — Elastic Waves & Elastodynamics #3 - Session Chair: Semyon Tsynkov**

| Time  | Title                                                                                                                     | Authors                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 10:30 | Boundary determination in anisotropic elasticity                                                                          | Joonas Ilmavirta, Mikko Salo, <b>Hjördis Schlüter</b> , Daniel Windisch |
| 11:00 | Seismic Reflection of Three-Dimensional Plane Waves in Functionally Graded Triclinic Media with Interfacial Imperfections | <b>Akanksha Srivastava</b> , Federico J Sabina                          |
| 11:30 | An inverse problem for estimating effective parameters in asymptotic elastodynamics with non-periodic roughness           | <b>Rodrigo Zelada</b> , Alexandre Imperiale, Philippe Moireau           |
| 12:00 | Exponential and harmonic regimes of anti-plane shear wave in a piezoelectric thin layer over half-space                   | <b>Anna Zemlyanova</b> , Gennadi Mikhasev, Arindam Nath, Jia Fei        |

## Lunch 12:30-14:00

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### Fri · Block B · 14:00 – 16:00

#### MB-9.A — Domain Decomposition & Preconditioning - Session Chair: Lise-Marie Imbert-Gerard

| Time  | Title                                                                                               | Authors                                              |
|-------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 14:00 | Substructured Optimized Schwarz Domain Decomposition Solver on Multiple GPUs for Helmholtz Problems | <b>R. Greffe</b> , A. Chabib, A. Modave, C. Geuzaine |
| 14:30 | Domain Decomposition for Elastic Waves in Spatial Networks                                          | <b>Joseph Holten</b> , Moritz Hauck                  |
| 15:00 | A Flux-Based Domain Decomposition Method for Time-Harmonic Wave Problems                            | <b>Nawfel Benatia</b> , Christophe Geuzaine          |

#### MB-9.B — Full Waveform Inversion & Geophysical Applications - Session Chair: Allison Malcolm

| Time  | Title                                                                                                       | Authors                                                |
|-------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 14:00 | Full Waveform Inversion Using Second Order Optimizers and Domain Decomposition with Spectral Coarse Space   | <b>Boris Martin</b> , Christophe Geuzaine              |
| 14:30 | An efficient numerical solver for non-periodic scattering problems in locally perturbed periodic structures | <b>Nasim Shafieeabyaneh</b> , Tilo Arens, Ruming Zhang |

**MB-9.C — Plasma & Mathematical Foundations - Session Chair: Yann-Meing Law**

| Time  | Title                                                                                                                                   | Authors                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 14:00 | Finite Element Formulation with Integral Dielectric Kernel for ICRH Plasma Heating on Non-Field-Aligned Meshes                          | <b>Jules Zaleski</b> , Bernard Reman, Philippe Lamalle, Christophe Geuzaine    |
| 14:30 | Geometry-dependent well-posedness and finite elements for shear Alfvén waves                                                            | <b>M. Renoldner</b> , A. Buffa, T. Miehl, M. Picasso                           |
| 15:00 | A discontinuous Galerkin approach for the radiative transfer approximation of high-frequency vibrational energy flow on curved surfaces | <b>Robert J. Lockett</b> , David J. Chappell                                   |
| 15:30 | Analysis and Simulation of 2D Maxwell's equations in Cold Plasma                                                                        | <b>Manaswinee Bezbaruah</b> <sup>1</sup> , Patrick Ciarlet, Maryna Kachanovska |

**MB-9.D — Time Integration & Nonlinear Waves - Session Chair: Tom Hagstrom**

| Time  | Title                                                                                                  | Authors                                                            |
|-------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 14:00 | A Posteriori Estimates for Space Discretizations of the Wave Equation with Dynamic Meshes              | Théophile Chaumont-Frelet, <b>Sumit Mahajan</b> , Martin Vohralík  |
| 14:30 | Preserving multiple conserved quantities for the Korteweg–de Vries equation                            | Andy T.S. Wan, <b>Tareq Uz Zaman</b>                               |
| 15:00 | Nonlinear Acoustic Model Equations resulting from Non-Relativistic and Relativistic Lagrangian Pairing | <b>Markus Scholle</b> , Philip H. Gaskell, Meriem Al Ahmadi Hammou |